

# DG32-LITE - Preliminary Datasheet

Lockstep RISC-V motor-control SoC on SkyWater sky130A, in a 64-pin QFN. Dual rv32imc cores with a fault comparator, 32 KB SRAM, 64 KB self-boot ROM and QSPI-XIP, a 3-phase motor peripheral set (PWM, DShot, encoder, CORDIC, on-die SAR ADC) and UART/SPI/I2C/GPIO. For firmware and hardware/PCB engineers.

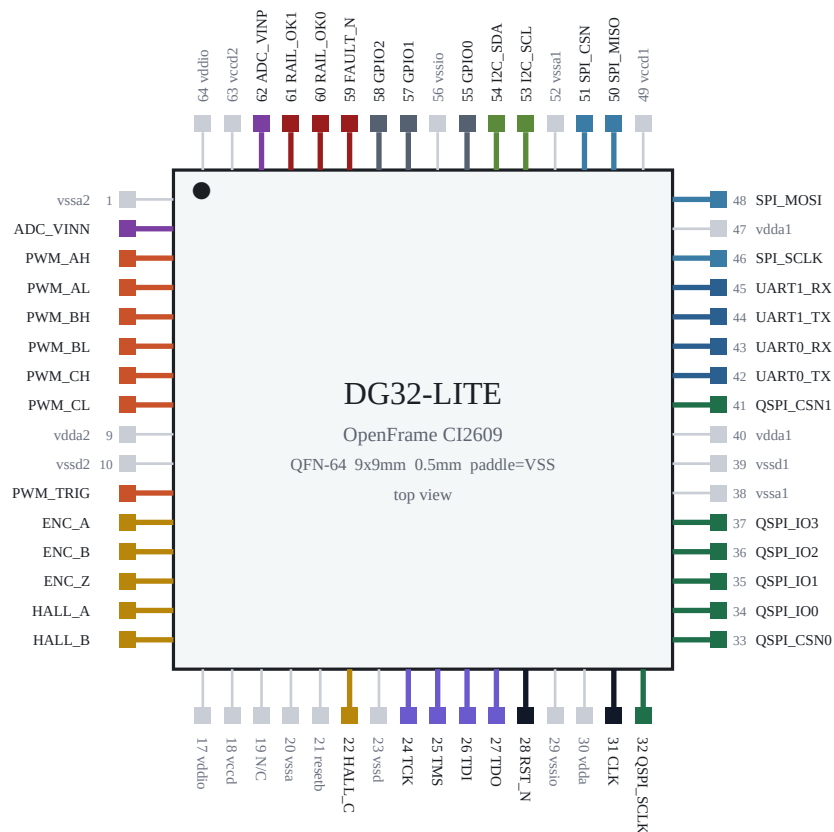
| Part      | Package        | Process        | Core                | Clock  |
|-----------|----------------|----------------|---------------------|--------|
| DG32-LITE | QFN-64, 9x9 mm | sky130A 130 nm | 2x rv32imc lockstep | 50 MHz |

**Preliminary, pre-silicon.** Functional and register detail is from the taped-out RTL and the hardened die. Electrical limits are design targets and sky130 process nominals pending first-silicon characterisation, and are marked where they are not yet measured. The flight-control core is frozen until silicon test.

## 1 Features

- Dual DGridRiscV rv32imc cores in lockstep with a cycle-by-cycle fault comparator and a sticky fault CSR.
- 32 KB on-chip SRAM (no ECC); 64 KB baked boot ROM; QSPI execute-in-place window for external NOR flash.
- 3-phase center-aligned complementary PWM with programmable dead-time and a hardware brake.
- 4-channel DShot ESC engine, pin-muxed onto the PWM output pads.
- Quadrature + Hall encoder with edge timestamps; CORDIC (sin/cos, atan2, magnitude) for field-oriented control.
- On-die 8-bit differential SAR ADC for phase-current feedback, PWM-triggered.
- Two UARTs, an SPI master, a 7-bit I2C master, 14-pin GPIO, two timers, a windowed watchdog and rail supervisor.
- Interrupt controller; standalone self-boot (no management core); SLVERR on any unmapped access.
- Single 50 MHz clock domain; whole die powered from a 1.8 V rail; 3.3 V I/O.

## 2 Pinout - 64-pin QFN (top view)



Pin map derived from the OpenFrame harness general pinout and the DG32-LITE pad plan (rev 2), awaiting ChipFoundry bond-diagram confirmation. All GPIO pads are 3.3 V CMOS; the two analog pads (VINP, VINN) carry the SAR inputs and must stay within 0 to 1.8 V.

## 3 Pin description

QFN pin 1 is at the upper-left; numbering runs counter-clockwise. User signals are the DG32 function on each OpenFrame user pad; power and ground pins are the harness pad-ring supplies. The exposed paddle is VSS.

| Pin | Signal           | Dir                              | Note                                                                                                                          | Pin | Signal           | Dir                | Note                                                                         |
|-----|------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----|------------------|--------------------|------------------------------------------------------------------------------|
| 1   | <b>vssa2</b>     | power                            | ground                                                                                                                        | 33  | <b>QSPI_CSN0</b> | out                | QSPI flash chip select 0 (boot flash)                                        |
| 2   | <b>ADC_VINN</b>  | analog                           | SAR ADC negative input, 0 to 1.8 V                                                                                            | 34  | <b>QSPI_IO0</b>  | bidir              | QSPI data 0                                                                  |
| 3   | <b>PWM_AH</b>    | out                              | phase A high-side gate                                                                                                        | 35  | <b>QSPI_IO1</b>  | bidir              | QSPI data 1                                                                  |
| 4   | <b>PWM_AL</b>    | out                              | phase A low-side gate                                                                                                         | 36  | <b>QSPI_IO2</b>  | bidir              | QSPI data 2                                                                  |
| 5   | <b>PWM_BH</b>    | out                              | phase B high-side gate                                                                                                        | 37  | <b>QSPI_IO3</b>  | bidir              | QSPI data 3                                                                  |
| 6   | <b>PWM_BL</b>    | out                              | phase B low-side gate                                                                                                         | 38  | <b>vssa1</b>     | power              | ground                                                                       |
| 7   | <b>PWM_CH</b>    | out                              | phase C high-side gate                                                                                                        | 39  | <b>vssd1</b>     | power              | ground (user area)                                                           |
| 8   | <b>PWM_CL</b>    | out                              | phase C low-side gate                                                                                                         | 40  | <b>vdda1</b>     | power              | 3.3 V user supply, unused by DG32-LITE — tie to 3.3 V                        |
| 9   | <b>vdda2</b>     | power                            | 3.3 V user supply, unused by DG32-LITE — tie to 3.3 V                                                                         | 41  | <b>QSPI_CSN1</b> | out                | QSPI chip select 1 (second device)                                           |
| 10  | <b>vssd2</b>     | power                            | ground                                                                                                                        | 42  | <b>UART0_TX</b>  | out                | console UART transmit; boot banner DG32" appears here"                       |
| 11  | <b>PWM_TRIG</b>  | out                              | ADC sample trigger (centre-aligned)                                                                                           | 43  | <b>UART0_RX</b>  | in                 | console UART receive                                                         |
| 12  | <b>ENC_A</b>     | in                               | quadrature encoder A                                                                                                          | 44  | <b>UART1_TX</b>  | out                | UART 1 transmit                                                              |
| 13  | <b>ENC_B</b>     | in                               | quadrature encoder B                                                                                                          | 45  | <b>UART1_RX</b>  | in                 | UART 1 receive                                                               |
| 14  | <b>ENC_Z</b>     | in                               | encoder index                                                                                                                 | 46  | <b>SPI_SCLK</b>  | out                | SPI master clock (external ADC / driver)                                     |
| 15  | <b>HALL_A</b>    | in                               | Hall sensor A                                                                                                                 | 47  | <b>vdda1</b>     | power              | 3.3 V user supply, unused by DG32-LITE — tie to 3.3 V                        |
| 16  | <b>HALL_B</b>    | in                               | Hall sensor B                                                                                                                 | 48  | <b>SPI_MOSI</b>  | out                | SPI master data out                                                          |
| 17  | <b>vddio</b>     | power                            | 3.3 V pad-ring / ESD supply — REQUIRED                                                                                        | 49  | <b>vccd1</b>     | power              | 1.8 V user-area supply: all DG32-LITE logic, SRAM and the SAR ADC — REQUIRED |
| 18  | <b>vccd</b>      | power                            | 1.8 V harness core supply — REQUIRED                                                                                          | 50  | <b>SPI_MISO</b>  | in                 | SPI master data in                                                           |
| 19  | <b>N/C</b>       | power                            | no connect                                                                                                                    | 51  | <b>SPI_CSN</b>   | out                | SPI master chip select                                                       |
| 20  | <b>vssa</b>      | power                            | ground                                                                                                                        | 52  | <b>vssa1</b>     | power              | ground                                                                       |
| 21  | <b>resetb</b>    | power                            | harness reset input, active low, 3.3 V — ANDed into the DG32-LITE core reset; may be left pulled up if RST_N (pin 28) is used | 53  | <b>I2C_SCL</b>   | bidir (open-drain) | I2C clock; external pull-up required                                         |
| 22  | <b>HALL_C</b>    | in                               | Hall sensor C                                                                                                                 | 54  | <b>I2C_SDA</b>   | bidir (open-drain) | I2C data; external pull-up required                                          |
| 23  | <b>vssd</b>      | power                            | ground                                                                                                                        | 55  | <b>GPIO0</b>     | bidir              | general-purpose I/O 0                                                        |
| 24  | <b>TCK</b>       | in (weak pull-down)              | JTAG clock                                                                                                                    | 56  | <b>vssio</b>     | power              | ground                                                                       |
| 25  | <b>TMS</b>       | in (weak pull-up)                | JTAG mode select                                                                                                              | 57  | <b>GPIO1</b>     | bidir              | general-purpose I/O 1                                                        |
| 26  | <b>TDI</b>       | in (weak pull-up)                | JTAG data in                                                                                                                  | 58  | <b>GPIO2</b>     | bidir              | general-purpose I/O 2                                                        |
| 27  | <b>TDO</b>       | out (driven only while shifting) | JTAG data out                                                                                                                 | 59  | <b>FAULT_N</b>   | out                | lockstep / supervisor fault, active low                                      |
| 28  | <b>RST_N</b>     | in                               | external reset, active low (ANDed on die with resetb and the harness POR)                                                     | 60  | <b>RAIL_OK0</b>  | in                 | rail supervisor input 0 (rail_ok[3:2] tied high on die)                      |
| 29  | <b>vssio</b>     | power                            | ground                                                                                                                        | 61  | <b>RAIL_OK1</b>  | in                 | rail supervisor input 1                                                      |
| 30  | <b>vdda</b>      | power                            | 3.3 V harness analog supply (POR) — REQUIRED                                                                                  | 62  | <b>ADC_VINP</b>  | analog             | SAR ADC positive input, 0 to 1.8 V (analog pad, digital buffer off)          |
| 31  | <b>CLK</b>       | in                               | system clock, 3.3 V CMOS, 48 MHz nominal (core clock)                                                                         | 63  | <b>vccd2</b>     | power              | 1.8 V user supply, unused by DG32-LITE — tie to 1.8 V                        |
| 32  | <b>QSPI_SCLK</b> | out                              | QSPI boot flash clock                                                                                                         | 64  | <b>vddio</b>     | power              | 3.3 V pad-ring / ESD supply — REQUIRED                                       |

### 3.1 Pins by function

| Function      | QFN pins                    |
|---------------|-----------------------------|
| Clock / reset | CLK=31, RST_N=28, resetb=21 |

| Function         | QFN pins                                                      |
|------------------|---------------------------------------------------------------|
| QSPI boot flash  | SCLK=32, CSN0=33, IO0=34, IO1=35, IO2=36, IO3=37, CSN1=41     |
| UART             | UART0 TX=42 RX=43; UART1 TX=44 RX=45                          |
| SPI master       | SCLK=46, MOSI=48, MISO=50, CSN=51                             |
| I2C              | SCL=53, SDA=54 (open-drain, external pull-ups)                |
| GPIO             | GPIO0=55, GPIO1=57, GPIO2=58                                  |
| Supervisor       | FAULT_N=59, RAIL_OK0=60, RAIL_OK1=61                          |
| SAR ADC (analog) | VINP=62, VINN=2 (0 to 1.8 V)                                  |
| Motor PWM        | AH=3, AL=4, BH=5, BL=6, CH=7, CL=8, TRIG=11                   |
| Position sensing | ENC_A=12, ENC_B=13, ENC_Z=14, HALL_A=15, HALL_B=16, HALL_C=22 |
| JTAG             | TCK=24, TMS=25, TDI=26, TDO=27                                |

## 4 Electrical characteristics

Pre-silicon. Recommended operating values are the sky130 process nominals for this die's supply domains; absolute-maximum values are conservative process limits, not device characterisation. First-silicon characterisation will replace this section.

### 4.1 Recommended operating conditions

| Parameter     | Sym           | Min  | Typ | Max  | Unit | Note                                                       |
|---------------|---------------|------|-----|------|------|------------------------------------------------------------|
| Core supply   | vccd1         | 1.71 | 1.8 | 1.89 | V    | sky130 core, +/-5%; powers all logic, SRAM and the SAR ADC |
| I/O supply    | vddio         | 3.0  | 3.3 | 3.6  | V    | 3.3 V CMOS pad ring / ESD                                  |
| Core clock    | CLK           | -    | 50  | ~55  | MHz  | fmax ~55-62 MHz measured post-route; 50 MHz design target  |
| ADC input     | VINP/VIN<br>N | 0    | -   | 1.8  | V    | differential, vccd1 domain                                 |
| Junction temp | TJ            | -    | 25  | -    | degC | power quoted at tt 25 C; range TBD on characterisation     |

### 4.2 Absolute maximum ratings (process limits, pre-characterisation)

| Parameter                  | Min  | Max         | Unit          |
|----------------------------|------|-------------|---------------|
| Core supply vccd1          | -0.3 | 1.95        | V             |
| I/O supply vddio           | -0.3 | 3.63        | V             |
| Voltage on any digital pad | -0.3 | vddio + 0.3 | V             |
| Analog pad voltage         | -0.3 | vccd1 + 0.3 | V             |
| Storage temperature        | -40  | 125         | degC (target) |

### 4.3 Power

Estimated total ~0.43 W at 50 MHz (OpenROAD vectorless, tt 25 C, 1.8 V). All dynamic and leakage current is drawn from vccd1; the other user supply domains are unused by DG32-LITE and are tied only to keep the pad ring's domain-order ESD structures biased.

## 5 Power supplies and sequencing

| Rail          | QFN pins                      | Voltage | Purpose                                               |
|---------------|-------------------------------|---------|-------------------------------------------------------|
| vddio         | 17, 64                        | 3.3 V   | pad-ring / ESD supply - REQUIRED                      |
| vccd          | 18                            | 1.8 V   | harness core supply - REQUIRED                        |
| vccd1         | 49                            | 1.8 V   | user-area supply: all logic, SRAM, SAR ADC - REQUIRED |
| vdda          | 30                            | 3.3 V   | harness analog / POR supply - REQUIRED                |
| vdda1         | 40, 47                        | 3.3 V   | user analog, unused - tie to 3.3 V                    |
| vdda2         | 9                             | 3.3 V   | user analog, unused - tie to 3.3 V                    |
| vccd2         | 63                            | 1.8 V   | user digital, unused - tie to 1.8 V                   |
| vss* / paddle | 1,10,20,23,29,39,52,56<br>... | 0 V     | all grounds and the exposed paddle                    |

**Sequencing.** Bring up 3.3 V (pad ring) before or together with 1.8 V, as on the ChipFoundry development board. All vss\* pins and the exposed paddle go to ground. DG32-LITE draws its whole current from vccd1 (1.8 V); tie the unused user rails (vdda1, vdda2, vccd2) to their nominal voltage so the ESD structures stay biased.

## 6 Clock, reset and boot

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### 6.1 Clock and reset

A single core clock enters on CLK (pin 31), 3.3 V CMOS, 50 MHz design target. There is no on-die PLL. Two reset inputs are ANDed on die: the OpenFrame harness resetb (pin 21, active low) and RST\_N (pin 28, active low); either may be used, pull the unused one high. The DG32-LITE die runs entirely from this one 50 MHz domain.

### 6.2 Boot flow

The die is standalone - no management core. At power-on the on-die 64 KB boot ROM runs with no external help:

- Emit the boot banner on UART0 (pin 42) at 115200 8N1 (baud divider fixed for 50 MHz).
- Read the QSPI NOR flash header through the 0x1 window (boot magic 0xD632\_B007, then length).
- Copy the flash image into SRAM over the QSPI quad-output-fast-read (0x6B, 1-1-4, 8 dummy cycles, SCLK = clk/2).
- Jump to the copied image and run from SRAM/ROM.
- A UART monitor (r/w/j/i commands) is available if the flash is blank, so a bare board is still inspectable.

**Instruction fetch is ROM-only.** The CPU fetch bus is wired to the boot ROM alone. The QSPI-XIP window and SRAM are reachable by loads and stores, not by instruction fetch - code does not execute in place from external NOR, it is copied to SRAM first. Firmware and linker scripts must place executable code in SRAM.

## 7 Memory map

Base-address decode from the top-level decoder: selector is addr[31:28]; the 0xD page sub-decodes addr[27:24], so each 0xD peripheral base is 0xDX00\_0000. Any unmapped access completes with SLVERR (it never hangs the bus).

| Base        | Block                | Note                                            |
|-------------|----------------------|-------------------------------------------------|
| 0x1000_0000 | QSPI-XIP read window | data read window (loads); NOT instruction fetch |
| 0x9000_0000 | SRAM                 | 32 KB main data memory                          |
| 0xC000_0000 | DShot ESC engine     | pin-muxed onto the PWM pads                     |
| 0xD000_0000 | UART0                | console / boot banner                           |
| 0xD100_0000 | UART1                | telemetry                                       |
| 0xD200_0000 | SPI master           |                                                 |
| 0xD300_0000 | GPIO                 | 14 pins on the die                              |
| 0xD400_0000 | TIMER0               |                                                 |
| 0xD500_0000 | TIMER1               |                                                 |
| 0xD600_0000 | Fault CSR            | poison + cause                                  |
| 0xD700_0000 | Supervisor / wWDt    |                                                 |
| 0xD800_0000 | PWM                  | 3-phase center-aligned                          |
| 0xD900_0000 | I2C master           |                                                 |
| 0xDA00_0000 | Encoder              | quadrature + Hall                               |
| 0xDB00_0000 | SAR ADC              | 8-bit differential                              |
| 0xDC00_0000 | DMA                  |                                                 |
| 0xDD00_0000 | SYSTCL               | device ID / clock gate                          |
| 0xDE00_0000 | CORDIC               | Q1.31                                           |
| 0xDF00_0000 | IRQ aggregator       |                                                 |

**Address constraint (firmware).** The CPU has a hardwired PMA whitelist: a load or store is legal only if addr[31] is set or the address is in 0x1xxx\_xxxx. That is why SRAM is at 0x9 and the peripherals at 0xC/0xD. An access outside the whitelist faults inside the CPU before the bus sees it.

## 8 Register map

Byte offsets within each block, extracted from the RTL and cross-checked against the AXI decode. Sticky bits clear as noted (W1C = write 1 to clear). Self-clearing start bits are marked in the block.

### UART0 / UART1 (0xD0 / 0xD1)

| Offset | Name          | R/W | Bits / meaning                                                                    |
|--------|---------------|-----|-----------------------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>   | R/W | [0]=enable; [31:16]=clkdiv (baud = clk_i/clkdiv)                                  |
| 0x04   | <b>TX</b>     | W   | [7:0]=byte to send (8N1); write back-pressures while tx_busy - never drops a byte |
| 0x08   | <b>RX</b>     | R   | [7:0]=received byte; read clears rx_valid                                         |
| 0x0C   | <b>STATUS</b> | R   | [0]=tx_busy; [1]=rx_valid                                                         |

8N1: 1 start + 8 data LSB-first + 1 stop. RX 2-flop synchronised. 115200 baud at 50 MHz = 434 clocks/bit.

### SPI master (0xD2)

| Offset | Name          | R/W | Bits / meaning                                                            |
|--------|---------------|-----|---------------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>   | R/W | [0]=enable; [1]=cpol; [2]=cpha; [9:4]=xfer_bits (8/16/32); [31:16]=clkdiv |
| 0x04   | <b>TX</b>     | W/R | write launches a full-duplex MSB-first transfer when enabled and not busy |
| 0x08   | <b>RX</b>     | R   | last received word                                                        |
| 0x0C   | <b>STATUS</b> | R   | [0]=busy                                                                  |
| 0x10   | <b>CS</b>     | R/W | [0]=cs_n level (reset 1 = deasserted)                                     |

SCLK idle level = CPOL. Transfer = 2 x xfer\_bits SCLK edges. Hardware master, 25 MHz, 0.64 us per 16-bit word.

### GPIO (0xD3)

| Offset | Name            | R/W   | Bits / meaning                                                       |
|--------|-----------------|-------|----------------------------------------------------------------------|
| 0x00   | <b>DATA_OUT</b> | R/W   | [N-1:0] value driven on pins whose OE bit is set                     |
| 0x04   | <b>OE</b>       | R/W   | [N-1:0] 1=drive, 0=high-Z input (reset 0 = all inputs); active-high  |
| 0x08   | <b>DATA_IN</b>  | R     | [N-1:0] 2-flop synchronised pin state; writes accepted and discarded |
| 0x0C   | <b>SET</b>      | W1S/R | write 1 sets that DATA_OUT bit; reads as DATA_OUT                    |
| 0x10   | <b>CLR</b>      | W1C/R | write 1 clears that DATA_OUT bit; reads as DATA_OUT                  |

SET/CLR are separate addresses onto one DATA\_OUT register (atomic single-pin toggling). N=14 on the die.

### TIMER0 / TIMER1 (0xD4 / 0xD5)

| Offset | Name          | R/W | Bits / meaning                                                             |
|--------|---------------|-----|----------------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>   | R/W | [0]=enable; [1]=auto_reload; [2]=irq_enable; write [8]=clear pending (W1C) |
| 0x04   | <b>RELOAD</b> | R/W | period-1 (fires every RELOAD+1 clocks)                                     |
| 0x08   | <b>COUNT</b>  | R   | current down-counter value                                                 |
| 0x0C   | <b>STATUS</b> | R   | [0]=pending (cleared via CTRL[8])                                          |

Down-counter; pulses at 0; irq = pending and irq\_enable. Enable-rising reloads count.

### Fault CSR (0xD6)

| Offset | Name          | R/W | Bits / meaning                                               |
|--------|---------------|-----|--------------------------------------------------------------|
| 0x0    | <b>STATUS</b> | R   | [0]=poison; [3:1]=cause. Sticky; cleared only by reset       |
| 0x4    | <b>INJECT</b> | W   | write MAGIC 0xDEAD_5AFE -> corrupt pulses 32 cycles. Reads 0 |
| 0x8    | <b>EVENTS</b> | R   | matched lockstep-compare count                               |
| 0xC    | <b>ID</b>     | R   | 0xD632_0001                                                  |

cause 001 = DATA\_MISMATCH. Sticky-first: holds the first cause across continued execution.

## Supervisor / wWDT (0xD7)

| Offset | Name            | R/W   | Bits / meaning                                                    |
|--------|-----------------|-------|-------------------------------------------------------------------|
| 0x00   | <b>ID</b>       | R     | 0x00D6_0006                                                       |
| 0x04   | <b>CTRL</b>     | R/W   | [0]=global_en; [1]=wdt_en; [7:4]=rail_en[3:0]                     |
| 0x08   | <b>DEGLITCH</b> | R/W   | [7:0] deglitch cycles (reset 8)                                   |
| 0x0C   | <b>HOLDOFF</b>  | R/W   | [15:0] reset-release holdoff (reset 1000)                         |
| 0x10   | <b>WDTMIN</b>   | R/W   | [15:0] watchdog window min (reset 100)                            |
| 0x14   | <b>WDTMAX</b>   | R/W   | [15:0] watchdog window max (reset 50000)                          |
| 0x18   | <b>KICK</b>     | W     | write 0xA5A5 = valid watchdog kick                                |
| 0x1C   | <b>STATUS</b>   | R/W1C | [0]=sys_rst_n; [1]=wdt_fault; [7:4]=rail_sticky; [15:8]=fault_cnt |

Windowed watchdog: a kick before WDTMIN or after WDTMAX both fault. Arms only when firmware sets global\_en.

## PWM (3-phase) (0xD8)

| Offset | Name             | R/W | Bits / meaning                                                       |
|--------|------------------|-----|----------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>      | R/W | [0]=EN                                                               |
| 0x04   | <b>PERIOD</b>    | R/W | counter top, 0..65535 (period = 2*(PERIOD+1) cycles, center-aligned) |
| 0x08   | <b>DEADTIME</b>  | R/W | dead-time cycles, all 3 legs (8-bit, max 255 = 5.1 us at 50 MHz)     |
| 0x0C   | <b>DUTY_A</b>    | R/W | phase A shadow duty, 0..PERIOD                                       |
| 0x10   | <b>DUTY_B</b>    | R/W | phase B shadow duty                                                  |
| 0x14   | <b>DUTY_C</b>    | R/W | phase C shadow duty                                                  |
| 0x18   | <b>STATUS</b>    | R   | [0]=BRAKED (sticky)                                                  |
| 0x1C   | <b>BRAKE_CLR</b> | W   | write [0]=1 to clear BRAKED once brake_i is low again                |

Shadow duties latch at cnt==0; sample\_trig pulses there. Async hardware brake forces all 6 outputs off in <=2 cycles.

## I2C master (0xD9)

| Offset | Name          | R/W   | Bits / meaning                                                                |
|--------|---------------|-------|-------------------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>   | R/W   | [0]=enable; [31:16]=clkdiv (SCL = clk_i/(4*(clkdiv+1))); reset 124 -> 100 kHz |
| 0x04   | <b>CMD</b>    | W/R   | [7:1]=7-bit addr; [0]=read/write; [15:8]=byte count 1..4; [16]=send STOP      |
| 0x08   | <b>TXD</b>    | R/W   | up to 4 bytes to send, LSB first on the wire                                  |
| 0x0C   | <b>RXD</b>    | R     | up to 4 bytes received, LSB first                                             |
| 0x10   | <b>STATUS</b> | R/W1C | [0]=busy; [1]=nack_addr; [2]=nack_data; [3]=arb_lost (write 1 to clear)       |

7-bit, 100/400 kHz, clock stretching, open-drain (external pull-ups). 2-byte read ~72 us at 400 kHz.

## Encoder (quad + Hall) (0xDA)

| Offset | Name                  | R/W   | Bits / meaning                                                   |
|--------|-----------------------|-------|------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>           | R/W   | [0]=EN (gates quad decode + Hall capture)                        |
| 0x04   | <b>POSITION</b>       | R/W   | signed 4x quadrature position; firmware may write to preset/zero |
| 0x08   | <b>INDEX_POSITION</b> | R     | POSITION latched at last Z rising edge                           |
| 0x0C   | <b>STATUS</b>         | R/W1C | [0]=INDEX_SEEN (sticky); [1]=DIR (live)                          |
| 0x10   | <b>TIMESTAMP</b>      | R     | free-running 32-bit cycle counter                                |
| 0x14   | <b>HALL_STATE</b>     | R     | [2:0]=synchronised {hall_c, hall_b, hall_a}                      |
| 0x18   | <b>HALL_TIMESTAMP</b> | R     | TIMESTAMP at last Hall transition                                |
| 0x1C   | <b>HALL_EVENT</b>     | R/W1C | [0]=NEW_EVENT (sticky)                                           |

Edge timestamps give velocity from time-between-edges, accurate at low speed.

## SAR ADC (0xDB)

| Offset | Name          | R/W | Bits / meaning                                                             |
|--------|---------------|-----|----------------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>   | R/W | [0]=enable; [1]=cal; [31:16]=clkdiv (reset 24 -> 50 MHz/25 = 2 MHz)        |
| 0x04   | <b>START</b>  | W   | any write starts one conversion when enabled and not busy                  |
| 0x08   | <b>RESULT</b> | R   | [7:0]=last conversion; [8]=valid (cleared on START)                        |
| 0x0C   | <b>STATUS</b> | R   | [0]=busy; [1]=valid; [2]=comp (raw comparator)                             |
| 0x10   | <b>TRIM</b>   | R/W | [4:0]=trim bits (reset 01111). NOTE: no effect on silicon - see block note |

8-bit differential SAR hard macro (1.8 V, on vccd1). ~200 kSa/s at the reset divider. The macro's trim/clkc/comp taps are walled inside its own metal and are left unrouted (routing them makes the die unroutable), so TRIM and CLKC have no effect on this silicon and STATUS.comp reads 0 - the ADC runs untrimmed. The register fields are kept for API stability.

## DShot ESC engine (0xC000\_0000)

| Offset | Name             | R/W | Bits / meaning                                     |
|--------|------------------|-----|----------------------------------------------------|
| 0x00   | <b>CTRL</b>      | W   | [0]=START (self-clearing; ignored while BUSY)      |
| 0x04   | <b>BITPERIOD</b> | R/W | cycles per bit slot (12-bit)                       |
| 0x08   | <b>T0H</b>       | R/W | high-time for logical 0                            |
| 0x0C   | <b>T1H</b>       | R/W | high-time for logical 1                            |
| 0x10   | <b>THROTTLE0</b> | R/W | [10:0]=throttle; [11]=telemetry request; channel 0 |
| 0x14   | <b>THROTTLE1</b> | R/W | channel 1                                          |
| 0x18   | <b>THROTTLE2</b> | R/W | channel 2                                          |
| 0x1C   | <b>THROTTLE3</b> | R/W | channel 3                                          |
| 0x20   | <b>STATUS</b>    | R   | [0]=BUSY                                           |

16-bit frame MSB-first: [15:5] throttle, [4] telemetry, [3:0] CRC4. All 4 channels share one bit-time counter.

## CORDIC (0xDE)

| Offset | Name          | R/W | Bits / meaning                                                |
|--------|---------------|-----|---------------------------------------------------------------|
| 0x00   | <b>CTRL</b>   | R/W | [0]=START (self-clearing); [1]=MODE (0=rotation, 1=vectoring) |
| 0x04   | <b>XIN</b>    | R/W | Q1.31 x input                                                 |
| 0x08   | <b>YIN</b>    | R/W | Q1.31 y input                                                 |
| 0x0C   | <b>ZIN</b>    | R/W | Q1.31 angle/pi                                                |
| 0x10   | <b>XOUT</b>   | R   | Q1.31 result x (rotation: cos; vectoring: magnitude/K)        |
| 0x14   | <b>YOUT</b>   | R   | Q1.31 result y (rotation: sin)                                |
| 0x18   | <b>ZOUT</b>   | R   | Q1.31 result z (vectoring: atan2/pi)                          |
| 0x1C   | <b>STATUS</b> | R   | [0]=BUSY                                                      |

N=20 iterations, ~53-58 cycles/op. K=0x4DBA76D4; vectoring magnitude is not gain-compensated by design.

## IRQ aggregator (0xDF)

| Offset | Name           | R/W   | Bits / meaning                                     |
|--------|----------------|-------|----------------------------------------------------|
| 0x00   | <b>PENDING</b> | R/W1C | [15:0] pending; write 1 to clear                   |
| 0x04   | <b>ENABLE</b>  | R/W   | [15:0] mask; external int = OR(PENDING and ENABLE) |
| 0x08   | <b>MSIP</b>    | R/W   | [0] -> machine software interrupt                  |
| 0x0C   | <b>RAW</b>     | R     | [15:0] source levels before edge detection         |
| 0x10   | <b>SWSET</b>   | W     | write 1 sets PENDING[7]                            |
| 0x14   | <b>ID</b>      | R     | 0x00D6_1249                                        |

Sources: 0 TMR0, 1 TMR1, 2 PWM trig, 3 ADC done, 4 DMA done, 5 I2C done, 6 FAULT, 7 SW, 8 ATTN (2DOM).

## DMA (0xDC)

| Offset | Name          | R/W   | Bits / meaning                                                    |
|--------|---------------|-------|-------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>   | R/W   | [0]=enable; [1]=start (self-clearing); [2]=src_incr; [3]=dst_incr |
| 0x04   | <b>SRC</b>    | R/W   | source address, word-aligned                                      |
| 0x08   | <b>DST</b>    | R/W   | destination address, word-aligned                                 |
| 0x0C   | <b>LEN</b>    | R/W   | transfer length in words (0 = none)                               |
| 0x10   | <b>STATUS</b> | R/W1C | [0]=busy; [1]=done (write 1 to clear); [2]=error                  |
| 0x14   | <b>COUNT</b>  | R     | words remaining                                                   |

Single-beat AXI master; the top arbitrates LSU-priority over DMA. ~7.1 cycles/word.

## 9 Board design guidance (hardware / PCB)

- Motor half-bridge gate driver on PWM\_AH..CL (pins 3-8); the outputs are complementary with programmable dead-time. Respect the driver's shoot-through spec; the die also has a hardware brake that forces all six low.
- Route PWM\_TRIG (pin 11) to nothing external unless you want the ADC-sample instant on a test point; it fires at the centre-aligned current-ripple null.
- Wire the encoder A/B/Z (pins 12-14) and Hall A/B/C (pins 15,16,22) directly; inputs are 2-flop synchronised and 3.3 V.
- SAR ADC inputs ADC\_VINP (62) and ADC\_VINN (2) are differential, 0 to 1.8 V - scale phase-current shunt amplifiers into that window and keep the pair matched and short.
- QSPI NOR boot flash on pins 32-37 (add CSN1=41 for a second device / PSRAM); a 3.3 V W25Q / S25FL works. Without a flash the ROM drops to the UART monitor.
- I2C\_SCL (53) and I2C\_SDA (54) are open-drain - the carrier must supply the pull-ups (typ 2.2-4.7 k to 3.3 V).
- Bring UART0 (42/43) to a USB-UART for the boot banner and console; UART1 (44/45) is free for telemetry.
- FAULT\_N (59) is an active-low fault output from the lockstep/supervisor - route it to the driver enable or a supervisor for a hardware trip.
- Provide 3.3 V and 1.8 V per Section 5; the 100 production parts need a carrier with the 64-pin QFN footprint (KiCad from chipfoundry/caravel\_board), gate drives, encoder, an optional external SPI ADC, and UART/JTAG headers.

**JTAG idles safely with no probe.** TMS (pin 25) and TDI (pin 26) are weakly pulled up on die, TCK (pin 24) weakly pulled down, so an unconnected JTAG header will not disturb the part.

## 10 Package

| Item           | Value                                   |
|----------------|-----------------------------------------|
| Package        | 64-pin QFN                              |
| Body           | 9.0 x 9.0 mm                            |
| Pitch          | 0.5 mm                                  |
| Exposed paddle | VSS (solder to ground plane)            |
| Pin 1          | upper-left, counter-clockwise numbering |
| Process        | SkyWater sky130A, 130 nm                |
| Shuttle        | chipIgnite CI2609                       |

Preliminary datasheet, pre-silicon. Register and pin detail from the taped-out RTL (CI2609 R2) and the QFN-64 pin map derived from the OpenFrame harness pinout, awaiting ChipFoundry bond-diagram confirmation. Electrical limits are design/process nominals pending first-silicon characterisation.

# DG32-2DOM - Preliminary Datasheet

The DG32-LITE motor-control SoC plus an INT8 attention accelerator on a second clock. Same lockstep core, boot path and motor peripheral set; adds a memory-mapped attention engine (0xE000\_0000) that runs on `clk_fast_i` and reaches memory through clock-domain-crossing bridges. For firmware and hardware/PCB engineers.

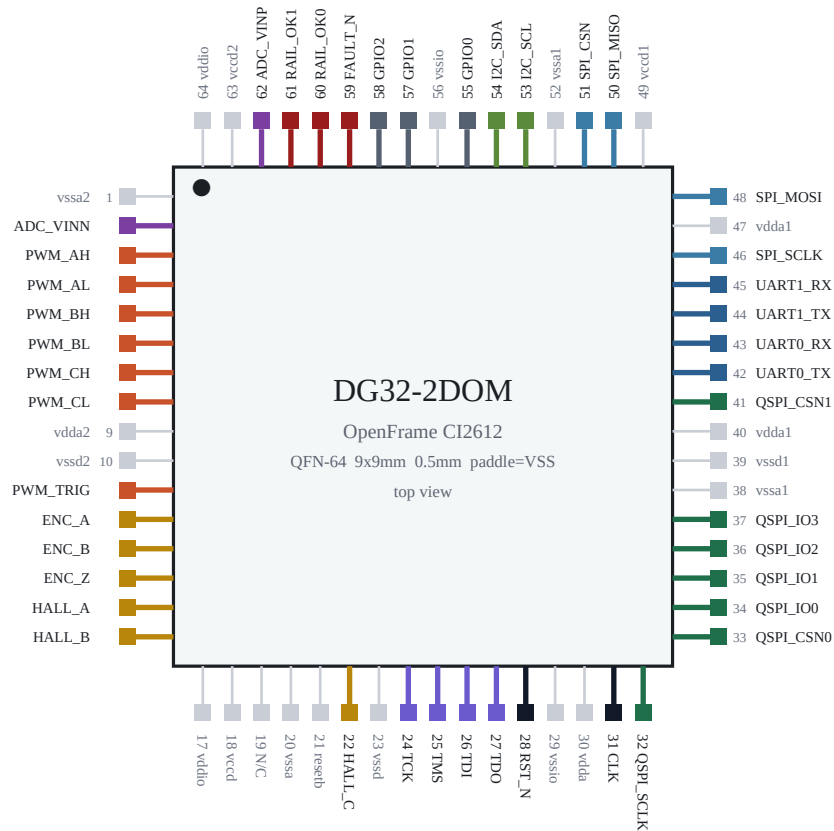
| Part      | Package        | Process        | Core                   | Clocks       |
|-----------|----------------|----------------|------------------------|--------------|
| DG32-2DOM | QFN-64, 9x9 mm | sky130A 130 nm | 2x rv32imc + INT8 attn | 50 / 114 MHz |

**Preliminary, pre-silicon.** Functional and register detail is from the taped-out RTL and the hardened die. Electrical limits are design targets and sky130 process nominals pending first-silicon characterisation, and are marked where they are not yet measured. The flight-control core is frozen until silicon test.

## 1 Features

- Everything in DG32-LITE: dual rv32imc lockstep core, 32 KB SRAM, 64 KB boot ROM, QSPI-XIP, and the full motor peripheral set (PWM, DShot, encoder, CORDIC, SAR ADC) plus UART/SPI/I2C/GPIO.
- INT8 attention accelerator (0xE000\_0000): one kick computes a band of query rows of one head end to end ( $QK^T$ , softmax/EXP, E.V, requantise to int8), bit-exact to the golden model.
- Attention datapath: shared LANES-wide MAC array, 256-entry u15 EXP table, 40-bit numerator (exact to  $NK=131072$ ), one reciprocal per row; ~3242 cycles/row analytic at LANES=16.
- Second clock domain: the engine runs on `clk_fast_i` (114 MHz) and crosses to the 50 MHz CPU domain through 2-flop CDC bridges, so it never stalls the frozen control core.
- Own die (3400 x 4500 um) with the attention K/V buffers on sky130 SRAM macros next to the SAR analog corner.

## 2 Pinout - 64-pin QFN (top view)



The external pinout is the same 64-pin QFN as DG32-LITE: the attention engine is internal (it talks to memory over the on-chip bus and to the CPU over AXI-lite), so it adds no user pads. The only added clock, `clk_fast_i`, is a review item - either derived on-die from CLK or brought in on a spare pad for the CI2612 die.

## 3 Pin description

QFN pin 1 is at the upper-left; numbering runs counter-clockwise. User signals are the DG32 function on each OpenFrame user pad; power and ground pins are the harness pad-ring supplies. The exposed paddle is VSS.

| Pin | Signal           | Dir                              | Note                                                                                                                          | Pin | Signal           | Dir                | Note                                                                         |
|-----|------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----|------------------|--------------------|------------------------------------------------------------------------------|
| 1   | <b>vssa2</b>     | power                            | ground                                                                                                                        | 33  | <b>QSPI_CSN0</b> | out                | QSPI flash chip select 0 (boot flash)                                        |
| 2   | <b>ADC_VINN</b>  | analog                           | SAR ADC negative input, 0 to 1.8 V                                                                                            | 34  | <b>QSPI_IO0</b>  | bidir              | QSPI data 0                                                                  |
| 3   | <b>PWM_AH</b>    | out                              | phase A high-side gate                                                                                                        | 35  | <b>QSPI_IO1</b>  | bidir              | QSPI data 1                                                                  |
| 4   | <b>PWM_AL</b>    | out                              | phase A low-side gate                                                                                                         | 36  | <b>QSPI_IO2</b>  | bidir              | QSPI data 2                                                                  |
| 5   | <b>PWM_BH</b>    | out                              | phase B high-side gate                                                                                                        | 37  | <b>QSPI_IO3</b>  | bidir              | QSPI data 3                                                                  |
| 6   | <b>PWM_BL</b>    | out                              | phase B low-side gate                                                                                                         | 38  | <b>vssa1</b>     | power              | ground                                                                       |
| 7   | <b>PWM_CH</b>    | out                              | phase C high-side gate                                                                                                        | 39  | <b>vssd1</b>     | power              | ground (user area)                                                           |
| 8   | <b>PWM_CL</b>    | out                              | phase C low-side gate                                                                                                         | 40  | <b>vdda1</b>     | power              | 3.3 V user supply, unused by DG32-LITE — tie to 3.3 V                        |
| 9   | <b>vdda2</b>     | power                            | 3.3 V user supply, unused by DG32-LITE — tie to 3.3 V                                                                         | 41  | <b>QSPI_CSN1</b> | out                | QSPI chip select 1 (second device)                                           |
| 10  | <b>vssd2</b>     | power                            | ground                                                                                                                        | 42  | <b>UART0_TX</b>  | out                | console UART transmit; boot banner DG32" appears here"                       |
| 11  | <b>PWM_TRIG</b>  | out                              | ADC sample trigger (centre-aligned)                                                                                           | 43  | <b>UART0_RX</b>  | in                 | console UART receive                                                         |
| 12  | <b>ENC_A</b>     | in                               | quadrature encoder A                                                                                                          | 44  | <b>UART1_TX</b>  | out                | UART 1 transmit                                                              |
| 13  | <b>ENC_B</b>     | in                               | quadrature encoder B                                                                                                          | 45  | <b>UART1_RX</b>  | in                 | UART 1 receive                                                               |
| 14  | <b>ENC_Z</b>     | in                               | encoder index                                                                                                                 | 46  | <b>SPI_SCLK</b>  | out                | SPI master clock (external ADC / driver)                                     |
| 15  | <b>HALL_A</b>    | in                               | Hall sensor A                                                                                                                 | 47  | <b>vdda1</b>     | power              | 3.3 V user supply, unused by DG32-LITE — tie to 3.3 V                        |
| 16  | <b>HALL_B</b>    | in                               | Hall sensor B                                                                                                                 | 48  | <b>SPI_MOSI</b>  | out                | SPI master data out                                                          |
| 17  | <b>vddio</b>     | power                            | 3.3 V pad-ring / ESD supply — REQUIRED                                                                                        | 49  | <b>vccd1</b>     | power              | 1.8 V user-area supply: all DG32-LITE logic, SRAM and the SAR ADC — REQUIRED |
| 18  | <b>vccd</b>      | power                            | 1.8 V harness core supply — REQUIRED                                                                                          | 50  | <b>SPI_MISO</b>  | in                 | SPI master data in                                                           |
| 19  | <b>N/C</b>       | power                            | no connect                                                                                                                    | 51  | <b>SPI_CSN</b>   | out                | SPI master chip select                                                       |
| 20  | <b>vssa</b>      | power                            | ground                                                                                                                        | 52  | <b>vssa1</b>     | power              | ground                                                                       |
| 21  | <b>resetb</b>    | power                            | harness reset input, active low, 3.3 V — ANDed into the DG32-LITE core reset; may be left pulled up if RST_N (pin 28) is used | 53  | <b>I2C_SCL</b>   | bidir (open-drain) | I2C clock; external pull-up required                                         |
| 22  | <b>HALL_C</b>    | in                               | Hall sensor C                                                                                                                 | 54  | <b>I2C_SDA</b>   | bidir (open-drain) | I2C data; external pull-up required                                          |
| 23  | <b>vssd</b>      | power                            | ground                                                                                                                        | 55  | <b>GPIO0</b>     | bidir              | general-purpose I/O 0                                                        |
| 24  | <b>TCK</b>       | in (weak pull-down)              | JTAG clock                                                                                                                    | 56  | <b>vssio</b>     | power              | ground                                                                       |
| 25  | <b>TMS</b>       | in (weak pull-up)                | JTAG mode select                                                                                                              | 57  | <b>GPIO1</b>     | bidir              | general-purpose I/O 1                                                        |
| 26  | <b>TDI</b>       | in (weak pull-up)                | JTAG data in                                                                                                                  | 58  | <b>GPIO2</b>     | bidir              | general-purpose I/O 2                                                        |
| 27  | <b>TDO</b>       | out (driven only while shifting) | JTAG data out                                                                                                                 | 59  | <b>FAULT_N</b>   | out                | lockstep / supervisor fault, active low                                      |
| 28  | <b>RST_N</b>     | in                               | external reset, active low (ANDed on die with resetb and the harness POR)                                                     | 60  | <b>RAIL_OK0</b>  | in                 | rail supervisor input 0 (rail_ok[3:2] tied high on die)                      |
| 29  | <b>vssio</b>     | power                            | ground                                                                                                                        | 61  | <b>RAIL_OK1</b>  | in                 | rail supervisor input 1                                                      |
| 30  | <b>vdda</b>      | power                            | 3.3 V harness analog supply (POR) — REQUIRED                                                                                  | 62  | <b>ADC_VINP</b>  | analog             | SAR ADC positive input, 0 to 1.8 V (analog pad, digital buffer off)          |
| 31  | <b>CLK</b>       | in                               | system clock, 3.3 V CMOS, 48 MHz nominal (core clock)                                                                         | 63  | <b>vccd2</b>     | power              | 1.8 V user supply, unused by DG32-LITE — tie to 1.8 V                        |
| 32  | <b>QSPI_SCLK</b> | out                              | QSPI boot flash clock                                                                                                         | 64  | <b>vddio</b>     | power              | 3.3 V pad-ring / ESD supply — REQUIRED                                       |

### 3.1 Pins by function

| Function      | QFN pins                    |
|---------------|-----------------------------|
| Clock / reset | CLK=31, RST_N=28, resetb=21 |

| Function             | QFN pins                                                      |
|----------------------|---------------------------------------------------------------|
| QSPI boot flash      | SCLK=32, CSN0=33, IO0=34, IO1=35, IO2=36, IO3=37, CSN1=41     |
| UART                 | UART0 TX=42 RX=43; UART1 TX=44 RX=45                          |
| SPI master           | SCLK=46, MOSI=48, MISO=50, CSN=51                             |
| I2C                  | SCL=53, SDA=54 (open-drain, external pull-ups)                |
| GPIO                 | GPIO0=55, GPIO1=57, GPIO2=58                                  |
| Supervisor           | FAULT_N=59, RAIL_OK0=60, RAIL_OK1=61                          |
| SAR ADC (analog)     | VINP=62, VINN=2 (0 to 1.8 V)                                  |
| Motor PWM            | AH=3, AL=4, BH=5, BL=6, CH=7, CL=8, TRIG=11                   |
| Position sensing     | ENC_A=12, ENC_B=13, ENC_Z=14, HALL_A=15, HALL_B=16, HALL_C=22 |
| JTAG                 | TCK=24, TMS=25, TDI=26, TDO=27                                |
| Attention (internal) | memory-mapped at 0xE000_0000; clk_fast_i is a review item     |

## 4 Electrical characteristics

Pre-silicon. Recommended operating values are the sky130 process nominals for this die's supply domains; absolute-maximum values are conservative process limits, not device characterisation. First-silicon characterisation will replace this section.

### 4.1 Recommended operating conditions

| Parameter     | Sym           | Min  | Typ | Max  | Unit | Note                                                       |
|---------------|---------------|------|-----|------|------|------------------------------------------------------------|
| Core supply   | vccd1         | 1.71 | 1.8 | 1.89 | V    | sky130 core, +/-5%; powers all logic, SRAM and the SAR ADC |
| I/O supply    | vddio         | 3.0  | 3.3 | 3.6  | V    | 3.3 V CMOS pad ring / ESD                                  |
| Core clock    | CLK           | -    | 50  | ~55  | MHz  | fmax ~55-62 MHz measured post-route; 50 MHz design target  |
| ADC input     | VINP/VIN<br>N | 0    | -   | 1.8  | V    | differential, vccd1 domain                                 |
| Junction temp | TJ            | -    | 25  | -    | degC | power quoted at tt 25 C; range TBD on characterisation     |

### 4.2 Absolute maximum ratings (process limits, pre-characterisation)

| Parameter                  | Min  | Max         | Unit          |
|----------------------------|------|-------------|---------------|
| Core supply vccd1          | -0.3 | 1.95        | V             |
| I/O supply vddio           | -0.3 | 3.63        | V             |
| Voltage on any digital pad | -0.3 | vddio + 0.3 | V             |
| Analog pad voltage         | -0.3 | vccd1 + 0.3 | V             |
| Storage temperature        | -40  | 125         | degC (target) |

### 4.3 Power

Estimated total ~0.43 W at 50 MHz (OpenROAD vectorless, tt 25 C, 1.8 V). All dynamic and leakage current is drawn from vccd1; the other user supply domains are unused by DG32-2DOM and are tied only to keep the pad ring's domain-order ESD structures biased.

## 5 Power supplies and sequencing

| Rail          | QFN pins                      | Voltage | Purpose                                               |
|---------------|-------------------------------|---------|-------------------------------------------------------|
| vddio         | 17, 64                        | 3.3 V   | pad-ring / ESD supply - REQUIRED                      |
| vccd          | 18                            | 1.8 V   | harness core supply - REQUIRED                        |
| vccd1         | 49                            | 1.8 V   | user-area supply: all logic, SRAM, SAR ADC - REQUIRED |
| vdda          | 30                            | 3.3 V   | harness analog / POR supply - REQUIRED                |
| vdda1         | 40, 47                        | 3.3 V   | user analog, unused - tie to 3.3 V                    |
| vdda2         | 9                             | 3.3 V   | user analog, unused - tie to 3.3 V                    |
| vccd2         | 63                            | 1.8 V   | user digital, unused - tie to 1.8 V                   |
| vss* / paddle | 1,10,20,23,29,39,52,56<br>... | 0 V     | all grounds and the exposed paddle                    |

**Sequencing.** Bring up 3.3 V (pad ring) before or together with 1.8 V, as on the ChipFoundry development board. All vss\* pins and the exposed paddle go to ground. DG32-2DOM draws its whole current from vccd1 (1.8 V); tie the unused user rails (vdda1, vdda2, vccd2) to their nominal voltage so the ESD structures stay biased.

## 6 Clock, reset and boot

---

### 6.1 Clock and reset

A single core clock enters on CLK (pin 31), 3.3 V CMOS, 50 MHz design target. There is no on-die PLL. Two reset inputs are ANDed on die: the OpenFrame harness resetb (pin 21, active low) and RST\_N (pin 28, active low); either may be used, pull the unused one high. DG32-2DOM adds a second clock, clk\_fast\_i (114 MHz), for the attention engine; the CPU and all peripherals stay in the 50 MHz domain and the two are bridged by CDC logic.

### 6.2 Boot flow

The die is standalone - no management core. At power-on the on-die 64 KB boot ROM runs with no external help:

- Emit the boot banner on UART0 (pin 42) at 115200 8N1 (baud divider fixed for 50 MHz).
- Read the QSPI NOR flash header through the 0x1 window (boot magic 0xD632\_B007, then length).
- Copy the flash image into SRAM over the QSPI quad-output-fast-read (0x6B, 1-1-4, 8 dummy cycles, SCLK = clk/2).
- Jump to the copied image and run from SRAM/ROM.
- A UART monitor (r/w/j/i commands) is available if the flash is blank, so a bare board is still inspectable.

**Instruction fetch is ROM-only.** The CPU fetch bus is wired to the boot ROM alone. The QSPI-XIP window and SRAM are reachable by loads and stores, not by instruction fetch - code does not execute in place from external NOR, it is copied to SRAM first. Firmware and linker scripts must place executable code in SRAM.

## 7 Memory map

Base-address decode from the top-level decoder: selector is `addr[31:28]`; the `0xD` page sub-decodes `addr[27:24]`, so each `0xD` peripheral base is `0xDX00_0000`. Any unmapped access completes with `SLVERR` (it never hangs the bus).

| Base                     | Block                | Note                                            |
|--------------------------|----------------------|-------------------------------------------------|
| <code>0x1000_0000</code> | QSPI-XIP read window | data read window (loads); NOT instruction fetch |
| <code>0x9000_0000</code> | SRAM                 | 32 KB main data memory                          |
| <code>0xC000_0000</code> | DShot ESC engine     | pin-muxed onto the PWM pads                     |
| <code>0xD000_0000</code> | UART0                | console / boot banner                           |
| <code>0xD100_0000</code> | UART1                | telemetry                                       |
| <code>0xD200_0000</code> | SPI master           |                                                 |
| <code>0xD300_0000</code> | GPIO                 | 14 pins on the die                              |
| <code>0xD400_0000</code> | TIMER0               |                                                 |
| <code>0xD500_0000</code> | TIMER1               |                                                 |
| <code>0xD600_0000</code> | Fault CSR            | poison + cause                                  |
| <code>0xD700_0000</code> | Supervisor / wWDT    |                                                 |
| <code>0xD800_0000</code> | PWM                  | 3-phase center-aligned                          |
| <code>0xD900_0000</code> | I2C master           |                                                 |
| <code>0xDA00_0000</code> | Encoder              | quadrature + Hall                               |
| <code>0xDB00_0000</code> | SAR ADC              | 8-bit differential                              |
| <code>0xDC00_0000</code> | DMA                  |                                                 |
| <code>0xDD00_0000</code> | SYSCTL               | device ID / clock gate                          |
| <code>0xDE00_0000</code> | CORDIC               | Q1.31                                           |
| <code>0xDF00_0000</code> | IRQ aggregator       |                                                 |
| <code>0xE000_0000</code> | INT8 attention       | DG32-2DOM only (INCLUDE_ATTN)                   |

**Address constraint (firmware).** The CPU has a hardwired PMA whitelist: a load or store is legal only if `addr[31]` is set or the address is in `0x1xxx_xxxx`. That is why SRAM is at `0x9` and the peripherals at `0xC/0xD/0xE`. An access outside the whitelist faults inside the CPU before the bus sees it.

## 8 Register map

Byte offsets within each block, extracted from the RTL and cross-checked against the AXI decode. Sticky bits clear as noted (W1C = write 1 to clear). Self-clearing start bits are marked in the block.

### UART0 / UART1 (0xD0 / 0xD1)

| Offset | Name          | R/W | Bits / meaning                                                                    |
|--------|---------------|-----|-----------------------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>   | R/W | [0]=enable; [31:16]=clkdiv (baud = clk_i/clkdiv)                                  |
| 0x04   | <b>TX</b>     | W   | [7:0]=byte to send (8N1); write back-pressures while tx_busy - never drops a byte |
| 0x08   | <b>RX</b>     | R   | [7:0]=received byte; read clears rx_valid                                         |
| 0x0C   | <b>STATUS</b> | R   | [0]=tx_busy; [1]=rx_valid                                                         |

8N1: 1 start + 8 data LSB-first + 1 stop. RX 2-flop synchronised. 115200 baud at 50 MHz = 434 clocks/bit.

### SPI master (0xD2)

| Offset | Name          | R/W | Bits / meaning                                                            |
|--------|---------------|-----|---------------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>   | R/W | [0]=enable; [1]=cpol; [2]=cpha; [9:4]=xfer_bits (8/16/32); [31:16]=clkdiv |
| 0x04   | <b>TX</b>     | W/R | write launches a full-duplex MSB-first transfer when enabled and not busy |
| 0x08   | <b>RX</b>     | R   | last received word                                                        |
| 0x0C   | <b>STATUS</b> | R   | [0]=busy                                                                  |
| 0x10   | <b>CS</b>     | R/W | [0]=cs_n level (reset 1 = deasserted)                                     |

SCLK idle level = CPOL. Transfer = 2 x xfer\_bits SCLK edges. Hardware master, 25 MHz, 0.64 us per 16-bit word.

### GPIO (0xD3)

| Offset | Name            | R/W   | Bits / meaning                                                       |
|--------|-----------------|-------|----------------------------------------------------------------------|
| 0x00   | <b>DATA_OUT</b> | R/W   | [N-1:0] value driven on pins whose OE bit is set                     |
| 0x04   | <b>OE</b>       | R/W   | [N-1:0] 1=drive, 0=high-Z input (reset 0 = all inputs); active-high  |
| 0x08   | <b>DATA_IN</b>  | R     | [N-1:0] 2-flop synchronised pin state; writes accepted and discarded |
| 0x0C   | <b>SET</b>      | W1S/R | write 1 sets that DATA_OUT bit; reads as DATA_OUT                    |
| 0x10   | <b>CLR</b>      | W1C/R | write 1 clears that DATA_OUT bit; reads as DATA_OUT                  |

SET/CLR are separate addresses onto one DATA\_OUT register (atomic single-pin toggling). N=14 on the die.

### TIMER0 / TIMER1 (0xD4 / 0xD5)

| Offset | Name          | R/W | Bits / meaning                                                             |
|--------|---------------|-----|----------------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>   | R/W | [0]=enable; [1]=auto_reload; [2]=irq_enable; write [8]=clear pending (W1C) |
| 0x04   | <b>RELOAD</b> | R/W | period-1 (fires every RELOAD+1 clocks)                                     |
| 0x08   | <b>COUNT</b>  | R   | current down-counter value                                                 |
| 0x0C   | <b>STATUS</b> | R   | [0]=pending (cleared via CTRL[8])                                          |

Down-counter; pulses at 0; irq = pending and irq\_enable. Enable-rising reloads count.

### Fault CSR (0xD6)

| Offset | Name          | R/W | Bits / meaning                                               |
|--------|---------------|-----|--------------------------------------------------------------|
| 0x0    | <b>STATUS</b> | R   | [0]=poison; [3:1]=cause. Sticky; cleared only by reset       |
| 0x4    | <b>INJECT</b> | W   | write MAGIC 0xDEAD_5AFE -> corrupt pulses 32 cycles. Reads 0 |
| 0x8    | <b>EVENTS</b> | R   | matched lockstep-compare count                               |
| 0xC    | <b>ID</b>     | R   | 0xD632_0001                                                  |

cause 001 = DATA\_MISMATCH. Sticky-first: holds the first cause across continued execution.

## Supervisor / wWDT (0xD7)

| Offset | Name            | R/W   | Bits / meaning                                                    |
|--------|-----------------|-------|-------------------------------------------------------------------|
| 0x00   | <b>ID</b>       | R     | 0x00D6_0006                                                       |
| 0x04   | <b>CTRL</b>     | R/W   | [0]=global_en; [1]=wdt_en; [7:4]=rail_en[3:0]                     |
| 0x08   | <b>DEGLITCH</b> | R/W   | [7:0] deglitch cycles (reset 8)                                   |
| 0x0C   | <b>HOLDOFF</b>  | R/W   | [15:0] reset-release holdoff (reset 1000)                         |
| 0x10   | <b>WDTMIN</b>   | R/W   | [15:0] watchdog window min (reset 100)                            |
| 0x14   | <b>WDTMAX</b>   | R/W   | [15:0] watchdog window max (reset 50000)                          |
| 0x18   | <b>KICK</b>     | W     | write 0xA5A5 = valid watchdog kick                                |
| 0x1C   | <b>STATUS</b>   | R/W1C | [0]=sys_rst_n; [1]=wdt_fault; [7:4]=rail_sticky; [15:8]=fault_cnt |

Windowed watchdog: a kick before WDTMIN or after WDTMAX both fault. Arms only when firmware sets global\_en.

## PWM (3-phase) (0xD8)

| Offset | Name             | R/W | Bits / meaning                                                       |
|--------|------------------|-----|----------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>      | R/W | [0]=EN                                                               |
| 0x04   | <b>PERIOD</b>    | R/W | counter top, 0..65535 (period = 2*(PERIOD+1) cycles, center-aligned) |
| 0x08   | <b>DEADTIME</b>  | R/W | dead-time cycles, all 3 legs (8-bit, max 255 = 5.1 us at 50 MHz)     |
| 0x0C   | <b>DUTY_A</b>    | R/W | phase A shadow duty, 0..PERIOD                                       |
| 0x10   | <b>DUTY_B</b>    | R/W | phase B shadow duty                                                  |
| 0x14   | <b>DUTY_C</b>    | R/W | phase C shadow duty                                                  |
| 0x18   | <b>STATUS</b>    | R   | [0]=BRAKED (sticky)                                                  |
| 0x1C   | <b>BRAKE_CLR</b> | W   | write [0]=1 to clear BRAKED once brake_i is low again                |

Shadow duties latch at cnt==0; sample\_trig pulses there. Async hardware brake forces all 6 outputs off in <=2 cycles.

## I2C master (0xD9)

| Offset | Name          | R/W   | Bits / meaning                                                                |
|--------|---------------|-------|-------------------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>   | R/W   | [0]=enable; [31:16]=clkdiv (SCL = clk_i/(4*(clkdiv+1))); reset 124 -> 100 kHz |
| 0x04   | <b>CMD</b>    | W/R   | [7:1]=7-bit addr; [0]=read/write; [15:8]=byte count 1..4; [16]=send STOP      |
| 0x08   | <b>TXD</b>    | R/W   | up to 4 bytes to send, LSB first on the wire                                  |
| 0x0C   | <b>RXD</b>    | R     | up to 4 bytes received, LSB first                                             |
| 0x10   | <b>STATUS</b> | R/W1C | [0]=busy; [1]=nack_addr; [2]=nack_data; [3]=arb_lost (write 1 to clear)       |

7-bit, 100/400 kHz, clock stretching, open-drain (external pull-ups). 2-byte read ~72 us at 400 kHz.

## Encoder (quad + Hall) (0xDA)

| Offset | Name                  | R/W   | Bits / meaning                                                   |
|--------|-----------------------|-------|------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>           | R/W   | [0]=EN (gates quad decode + Hall capture)                        |
| 0x04   | <b>POSITION</b>       | R/W   | signed 4x quadrature position; firmware may write to preset/zero |
| 0x08   | <b>INDEX_POSITION</b> | R     | POSITION latched at last Z rising edge                           |
| 0x0C   | <b>STATUS</b>         | R/W1C | [0]=INDEX_SEEN (sticky); [1]=DIR (live)                          |
| 0x10   | <b>TIMESTAMP</b>      | R     | free-running 32-bit cycle counter                                |
| 0x14   | <b>HALL_STATE</b>     | R     | [2:0]=synchronised {hall_c, hall_b, hall_a}                      |
| 0x18   | <b>HALL_TIMESTAMP</b> | R     | TIMESTAMP at last Hall transition                                |
| 0x1C   | <b>HALL_EVENT</b>     | R/W1C | [0]=NEW_EVENT (sticky)                                           |

Edge timestamps give velocity from time-between-edges, accurate at low speed.

## SAR ADC (0xDB)

| Offset | Name          | R/W | Bits / meaning                                                      |
|--------|---------------|-----|---------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>   | R/W | [0]=enable; [1]=cal; [31:16]=clkdiv (reset 24 -> 50 MHz/25 = 2 MHz) |
| 0x04   | <b>START</b>  | W   | any write starts one conversion when enabled and not busy           |
| 0x08   | <b>RESULT</b> | R   | [7:0]=last conversion; [8]=valid (cleared on START)                 |
| 0x0C   | <b>STATUS</b> | R   | [0]=busy; [1]=valid; [2]=comp (raw comparator)                      |
| 0x10   | <b>TRIM</b>   | R/W | [4:0]=trim bits; trimb driven as complement (reset 01111)           |

8-bit differential SAR hard macro (1.8 V). ~200 kSa/s at reset divider; per-die trim is a bring-up step.

## DShot ESC engine (0xC000\_0000)

| Offset | Name             | R/W | Bits / meaning                                     |
|--------|------------------|-----|----------------------------------------------------|
| 0x00   | <b>CTRL</b>      | W   | [0]=START (self-clearing; ignored while BUSY)      |
| 0x04   | <b>BITPERIOD</b> | R/W | cycles per bit slot (12-bit)                       |
| 0x08   | <b>T0H</b>       | R/W | high-time for logical 0                            |
| 0x0C   | <b>T1H</b>       | R/W | high-time for logical 1                            |
| 0x10   | <b>THROTTLE0</b> | R/W | [10:0]=throttle; [11]=telemetry request; channel 0 |
| 0x14   | <b>THROTTLE1</b> | R/W | channel 1                                          |
| 0x18   | <b>THROTTLE2</b> | R/W | channel 2                                          |
| 0x1C   | <b>THROTTLE3</b> | R/W | channel 3                                          |
| 0x20   | <b>STATUS</b>    | R   | [0]=BUSY                                           |

16-bit frame MSB-first: [15:5] throttle, [4] telemetry, [3:0] CRC4. All 4 channels share one bit-time counter.

## CORDIC (0xDE)

| Offset | Name          | R/W | Bits / meaning                                                |
|--------|---------------|-----|---------------------------------------------------------------|
| 0x00   | <b>CTRL</b>   | R/W | [0]=START (self-clearing); [1]=MODE (0=rotation, 1=vectoring) |
| 0x04   | <b>XIN</b>    | R/W | Q1.31 x input                                                 |
| 0x08   | <b>YIN</b>    | R/W | Q1.31 y input                                                 |
| 0x0C   | <b>ZIN</b>    | R/W | Q1.31 angle/pi                                                |
| 0x10   | <b>XOUT</b>   | R   | Q1.31 result x (rotation: cos; vectoring: magnitude/K)        |
| 0x14   | <b>YOUT</b>   | R   | Q1.31 result y (rotation: sin)                                |
| 0x18   | <b>ZOUT</b>   | R   | Q1.31 result z (vectoring: atan2/pi)                          |
| 0x1C   | <b>STATUS</b> | R   | [0]=BUSY                                                      |

N=20 iterations, ~53-58 cycles/op. K=0x4DBA76D4; vectoring magnitude is not gain-compensated by design.

## IRQ aggregator (0xDF)

| Offset | Name           | R/W   | Bits / meaning                                     |
|--------|----------------|-------|----------------------------------------------------|
| 0x00   | <b>PENDING</b> | R/W1C | [15:0] pending; write 1 to clear                   |
| 0x04   | <b>ENABLE</b>  | R/W   | [15:0] mask; external int = OR(PENDING and ENABLE) |
| 0x08   | <b>MSIP</b>    | R/W   | [0] -> machine software interrupt                  |
| 0x0C   | <b>RAW</b>     | R     | [15:0] source levels before edge detection         |
| 0x10   | <b>SWSET</b>   | W     | write 1 sets PENDING[7]                            |
| 0x14   | <b>ID</b>      | R     | 0x00D6_1249                                        |

Sources: 0 TMR0, 1 TMR1, 2 PWM trig, 3 ADC done, 4 DMA done, 5 I2C done, 6 FAULT, 7 SW, 8 ATTN (2DOM).

## DMA (0xDC)

| Offset | Name          | R/W   | Bits / meaning                                                    |
|--------|---------------|-------|-------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>   | R/W   | [0]=enable; [1]=start (self-clearing); [2]=src_incr; [3]=dst_incr |
| 0x04   | <b>SRC</b>    | R/W   | source address, word-aligned                                      |
| 0x08   | <b>DST</b>    | R/W   | destination address, word-aligned                                 |
| 0x0C   | <b>LEN</b>    | R/W   | transfer length in words (0 = none)                               |
| 0x10   | <b>STATUS</b> | R/W1C | [0]=busy; [1]=done (write 1 to clear); [2]=error                  |
| 0x14   | <b>COUNT</b>  | R     | words remaining                                                   |

Single-beat AXI master; the top arbitrates LSU-priority over DMA. ~7.1 cycles/word.

## INT8 attention engine (0xE000\_0000) - programming (writes)

| Offset | Name           | R/W | Bits / meaning                                                              |
|--------|----------------|-----|-----------------------------------------------------------------------------|
| 0x00   | <b>CTRL</b>    | W   | [0]=start (self-clearing); [1]=SKIP_LOAD (reuse EXP/K/V from previous kick) |
| 0x04   | <b>A_BASE</b>  | W   | Q base byte address                                                         |
| 0x08   | <b>A_PITCH</b> | W   | Q position pitch (bytes)                                                    |
| 0x0C   | <b>B_BASE</b>  | W   | K base byte address                                                         |
| 0x10   | <b>B_PITCH</b> | W   | K position pitch (bytes)                                                    |
| 0x14   | <b>D_BASE</b>  | W   | O (output) base byte address                                                |
| 0x18   | <b>D_PITCH</b> | W   | O position pitch (bytes)                                                    |
| 0x1C   | <b>M</b>       | W   | MROWS: query rows this kick (1..65535)                                      |
| 0x20   | <b>N</b>       | W   | NK: keys (1..MAXK=400)                                                      |
| 0x24   | <b>K</b>       | W   | KD: Q/K head dim in bytes (mult of LANES and 4)                             |
| 0x28   | <b>SHIFT</b>   | W   | EXP index right-shift (6-bit)                                               |
| 0x2C   | <b>C_BASE</b>  | W   | V base byte address                                                         |
| 0x30   | <b>C_PITCH</b> | W   | V position pitch (bytes)                                                    |
| 0x34   | <b>DV</b>      | W   | output channels/head in bytes (mult of LANES and 4)                         |
| 0x38   | <b>E_BASE</b>  | W   | EXP table base (256 u15 entries)                                            |
| 0x3C   | <b>RATIO</b>   | W   | u32 16.16 window constant; rq = floor((RATIO<<16)/Z)                        |

## INT8 attention engine (0xE000\_0000) - status (reads)

| Offset | Name          | R/W | Bits / meaning                                                                           |
|--------|---------------|-----|------------------------------------------------------------------------------------------|
| 0x00   | <b>STATUS</b> | R   | [0]=busy; [1]=done; [2]=Z==0 seen; [3]=rq overflow; [4]=EXP clamp; [5]=geometry rejected |
| 0x40   | <b>Z</b>      | R   | Z of the last completed row                                                              |
| 0x44   | <b>rq</b>     | R   | rq of the last completed row                                                             |

**Programming a kick (firmware).** Write the Q/K/V/O base and pitch, the geometry (M rows, N keys, KD, DV), SHIFT, the EXP table base and RATIO, then set CTRL[0]=start. Poll STATUS[0]=busy to 0; STATUS[1]=done. SKIP\_LOAD (CTRL[1]) reuses the K/V/EXP buffers from the previous kick. The engine runs on clk\_fast\_i (114 MHz) and reaches memory through the CDC bridges, so it does not stall the 50 MHz CPU.

## 9 Board design guidance (hardware / PCB)

- Motor half-bridge gate driver on PWM\_AH..CL (pins 3-8); the outputs are complementary with programmable dead-time. Respect the driver's shoot-through spec; the die also has a hardware brake that forces all six low.
- Route PWM\_TRIG (pin 11) to nothing external unless you want the ADC-sample instant on a test point; it fires at the centre-aligned current-ripple null.
- Wire the encoder A/B/Z (pins 12-14) and Hall A/B/C (pins 15,16,22) directly; inputs are 2-flop synchronised and 3.3 V.
- SAR ADC inputs ADC\_VINP (62) and ADC\_VINN (2) are differential, 0 to 1.8 V - scale phase-current shunt amplifiers into that window and keep the pair matched and short.
- QSPI NOR boot flash on pins 32-37 (add CSN1=41 for a second device / PSRAM); a 3.3 V W25Q / S25FL works. Without a flash the ROM drops to the UART monitor.
- I2C\_SCL (53) and I2C\_SDA (54) are open-drain - the carrier must supply the pull-ups (typ 2.2-4.7 k to 3.3 V).
- Bring UART0 (42/43) to a USB-UART for the boot banner and console; UART1 (44/45) is free for telemetry.
- FAULT\_N (59) is an active-low fault output from the lockstep/supervisor - route it to the driver enable or a supervisor for a hardware trip.
- Provide 3.3 V and 1.8 V per Section 5; the 100 production parts need a carrier with the 64-pin QFN footprint (KiCad from chipfoundry/caravel\_board), gate drives, encoder, an optional external SPI ADC, and UART/JTAG headers.
- The attention engine adds no board signals; budget for clk\_fast\_i only if the CI2612 die brings it out to a pad rather than deriving it on-die.

**JTAG idles safely with no probe.** TMS (pin 25) and TDI (pin 26) are weakly pulled up on die, TCK (pin 24) weakly pulled down, so an unconnected JTAG header will not disturb the part.

## 10 Package

| Item           | Value                                   |
|----------------|-----------------------------------------|
| Package        | 64-pin QFN                              |
| Body           | 9.0 x 9.0 mm                            |
| Pitch          | 0.5 mm                                  |
| Exposed paddle | VSS (solder to ground plane)            |
| Pin 1          | upper-left, counter-clockwise numbering |
| Process        | SkyWater sky130A, 130 nm                |
| Shuttle        | chipIgnite CI2612                       |

Preliminary datasheet, pre-silicon. Register and pin detail from the taped-out RTL (CI2609 R2) and the QFN-64 pin map derived from the OpenFrame harness pinout, awaiting ChipFoundry bond-diagram confirmation. Electrical limits are design/process nominals pending first-silicon characterisation. DG32-2DOM adds the INT8 attention engine and a second (114 MHz) clock domain; its A3 die closed with core +0.30 ns at 50 MHz and clk\_fast +0.19 ns at 114 MHz.